



Particle Motion: Velocity and Acceleration from Position Functions

Calculus Worksheet · Grade 11-12

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Learning Objectives

- Compute velocity as the first derivative of a position function
- Compute acceleration as the second derivative of a position function
- Determine when a particle is at rest by solving $v(t) = 0$

For each position function $s(t)$, find the requested velocity, acceleration, or time(s) at rest, showing all derivative work.

1. Given the position function below (s in meters, t in seconds), find the velocity after 3 seconds.

$$s(t) = 2t^3 - 7t^2 + 4t + 1$$

Answer: _____

2. Using the same position function, find the acceleration after 1 second.

$$s(t) = 2t^3 - 7t^2 + 4t + 1$$

Answer: _____

3. Using the same position function, determine the time(s) when the particle is at rest.

$$s(t) = 2t^3 - 7t^2 + 4t + 1$$

Answer: _____

4. Find the velocity function and evaluate it at $t = 2$ seconds for the position function below.

$$s(t) = t^3 - 6t^2 + 9t + 5$$

Answer: _____

5. Find the acceleration after 4 seconds for the position function below.

$$s(t) = t^3 - 6t^2 + 9t + 5$$

Answer: _____

6. Determine the time(s) when the particle described below is at rest.

$$s(t) = t^3 - 6t^2 + 9t + 5$$

Answer: _____

7. Find the velocity after 5 seconds for the position function below.

$$s(t) = 4t^2 - 3t + 7$$

Answer: _____



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8. Find the acceleration function and its value at $t = 2$ seconds for the position function below.

$$s(t) = 4t^2 - 3t + 7$$

Answer: _____

9. Determine the time when the particle below is at rest.

$$s(t) = t^2 - 10t + 12$$

Answer: _____

10. For the position function below, find both the velocity and acceleration at $t = 1$ second.

$$s(t) = t^4 - 4t^3 + 6t$$

Answer: _____





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ANSWER KEY & SOLUTIONS

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Students should recall the power rule and zero product property; remind them that 'at rest' means $v(t) = 0$, not $s(t) = 0$.

Solutions

1. Given the position function below (s in meters, t in seconds), find the velocity after 3 seconds.

$$s(t) = 2t^3 - 7t^2 + 4t + 1$$

→ Differentiate $s(t)$ using the power rule to obtain $v(t) = 6t^2 - 14t + 4$.

→ Substitute $t = 3$: $v(3) = 6(9) - 14(3) + 4 = 54 - 42 + 4 = 16$ m/s.

Answer: $v(3) = 16$ m/s

2. Using the same position function, find the acceleration after 1 second.

$$s(t) = 2t^3 - 7t^2 + 4t + 1$$

→ Differentiate $v(t) = 6t^2 - 14t + 4$ to get $a(t) = 12t - 14$.

→ Substitute $t = 1$: $a(1) = 12(1) - 14 = -2$ m/s².

Answer: $a(1) = -2$ m/s²

3. Using the same position function, determine the time(s) when the particle is at rest.

$$s(t) = 2t^3 - 7t^2 + 4t + 1$$

→ Set $v(t) = 0$: $6t^2 - 14t + 4 = 0$, which factors as $(3t - 1)(t - 2) = 0$ after dividing by 2.

→ Apply the zero product property: $t = 1/3$ second and $t = 2$ seconds.

Answer: $t = \frac{1}{3}$ s, $t = 2$ s

4. Find the velocity function and evaluate it at $t = 2$ seconds for the position function below.

$$s(t) = t^3 - 6t^2 + 9t + 5$$

→ Differentiate to get $v(t) = 3t^2 - 12t + 9$.

→ Substitute $t = 2$: $v(2) = 3(4) - 12(2) + 9 = 12 - 24 + 9 = -3$ m/s.

Answer: $v(2) = -3$ m/s

5. Find the acceleration after 4 seconds for the position function below.

$$s(t) = t^3 - 6t^2 + 9t + 5$$

→ Differentiate $v(t) = 3t^2 - 12t + 9$ to get $a(t) = 6t - 12$.

→ Substitute $t = 4$: $a(4) = 6(4) - 12 = 12$ m/s².

Answer: $a(4) = 12$ m/s²

6. Determine the time(s) when the particle described below is at rest.

$$s(t) = t^3 - 6t^2 + 9t + 5$$

→ Set $v(t) = 3t^2 - 12t + 9 = 0$ and divide by 3 to get $t^2 - 4t + 3 = 0$.

→ Factor as $(t - 1)(t - 3) = 0$, giving $t = 1$ second and $t = 3$ seconds.

Answer: $t = 1$ s, $t = 3$ s



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7. Find the velocity after 5 seconds for the position function below.

$$s(t) = 4t^2 - 3t + 7$$

→ Differentiate to get $v(t) = 8t - 3$.

→ Substitute $t = 5$: $v(5) = 8(5) - 3 = 37 \text{ m/s}$.

Answer: $v(5) = 37 \text{ m/s}$

8. Find the acceleration function and its value at $t = 2$ seconds for the position function below.

$$s(t) = 4t^2 - 3t + 7$$

→ Differentiate $v(t) = 8t - 3$ to get $a(t) = 8$.

→ Since $a(t)$ is constant, $a(2) = 8 \text{ m/s}^2$.

Answer: $a(2) = 8 \text{ m/s}^2$

9. Determine the time when the particle below is at rest.

$$s(t) = t^2 - 10t + 12$$

→ Differentiate to get $v(t) = 2t - 10$.

→ Set $2t - 10 = 0$ and solve to obtain $t = 5$ seconds.

Answer: $t = 5 \text{ s}$

10. For the position function below, find both the velocity and acceleration at $t = 1$ second.

$$s(t) = t^4 - 4t^3 + 6t$$

→ Differentiate $s(t)$ to get $v(t) = 4t^3 - 12t^2 + 6$, then substitute $t = 1$ to find $v(1) = 4 - 12 + 6 = -2 \text{ m/s}$.

→ Differentiate again to get $a(t) = 12t^2 - 24t$, then substitute $t = 1$ to find $a(1) = 12 - 24 = -12 \text{ m/s}^2$.

Answer: $v(1) = -2 \text{ m/s}$, $a(1) = -12 \text{ m/s}^2$

